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A Blockchain-Based Implementation for Drug Production Transparency on Pharmaceutical Supply Chain

Abstract

The complexity of healthcare supply chains has long been associated with inefficiencies, lack of transparency, and the circulation of counterfeit drugs, which directly threaten patient safety and pharmaceutical reliability. This study introduces MediSync, a blockchain-based framework developed with Hyperledger Fabric to enhance transparency, traceability, and data integrity in pharmaceutical supply chains. Employing a hybrid architecture that combines on-chain hashing with off-chain storage, the system was designed to mitigate risks of data leakage while maintaining operational efficiency. The implementation was validated through network deployment, smart contract execution, decentralized governance testing, and a web-based interface enabling stakeholders to track drugs from production to end-user. Performance evaluation using Apache JMeter revealed a stark contrast between blockchain and conventional databases: while the off-chain MySQL system achieved higher throughput (101.3 tx/s) and lower latency (12 ms), the blockchain recorded significantly lower throughput (7.7 tx/s) and higher latency (1256 ms), reflecting inherent limitations of consensus mechanisms in high-load scenarios. Nonetheless, the blockchain demonstrated robustness in transaction integrity, public verification through QR codes, and prevention of data tampering, ensuring a reliable audit trail across stakeholders. Critically, the findings suggest that blockchain is more suitable for applications prioritizing integrity and authenticity rather than real-time responsiveness, with hybrid approaches offering a more balanced solution for large-scale deployment.

1. INTRODUCTION

The advancement of information and communication technology has catalyzed significant transformations across various sectors, including healthcare and pharmaceuticals. Technology facilitates health services by automating manual processes, improving the accuracy of medical data, and accelerating information exchange, thereby reducing errors and supporting faster, more informed decision-making (Amir & Taiwo, 2024). Information sharing and traceability serve as mediators in the connection with digital technologies and sustainable supply chain performance (Ma et al., 2023). In the pharmaceutical supply chain, which involves stakeholders such as manufacturers, Pharmaceutical Wholesale Distributors, and pharmacies, technology is vital for ensuring that medicines reach consumers with their quality preserved (Agarwal et al., 2022).

The healthcare supply chain is a complex network with several weaknesses. One consequence of supply chain complexity is the lack of consistent and standardized data that can identify, or at least signal, supply and quality issues to regulators and healthcare providers. In the absence of such data, policymakers do not have sufficient information to make investments to improve supply chain resilience, regulators are unable to identify raw material and medicine supply issues and compliance gaps, and healthcare providers are surprised by

shortages of critical medicines (United States Pharmacopeia (USP), 2020). Reporting mechanisms for drug/tool availability in many countries are inconsistent or not real-time, resulting in delayed responses and difficulties in redistribution. The evaluation of reporting systems indicated accessibility and sustainability issues of the MSRS (Medicines Shortages Reporting Systems) (Vann Yaroson et al., 2024). Despite these advancements, the application of technology within the pharmaceutical supply chain remains insufficient. Many existing systems rely on centralized, traditional approaches that cannot guarantee complete data transparency and security. This is a critical gap, as the pharmaceutical sector faces increasing demands for patient-oriented services that prioritize drug quality and patient safety (Gu, 2024). Various obstacles in the pharmaceutical supply chain contribute to this problem, including delays in drug delivery, the risk of drug counterfeiting (which includes the circulation of illegal drugs or drugs with substandard ingredients), and errors in drug distribution. Although many drug distribution systems adhere to safety guidelines, several global studies have shown that drug delivery delays and doubts about drug quality still occur, contributing to the risk of preventable, medication-related losses (WHO, 2024). According to the Health Research Funding Organization, 30% of medicines sold in developing countries are counterfeit. A recent study by the World Health Organization (WHO) found that children are the most frequent victims of counterfeit medicines, one of the leading causes of death in developing countries. Counterfeit medicines endanger human lives and cause significant financial losses for the pharmaceutical industry (Padma & Ramaiah, 2024).

The root cause of these problems lies in the lack of transparency and data security inherent in current centralized systems. These systems are susceptible to data manipulation, manual input errors, and flawed communication between parties (Dhingra et al., 2024). Information regarding a drug's origin or storage conditions often lacks proper, immutable validation. Furthermore, the absence of real-time tracking from the manufacturer to the pharmacy makes it difficult to monitor inventory and verify product authenticity, thus hindering effective prevention of counterfeiting (Gruchmann et al., 2023).

The urgency to address these issues is driven by their direct impact on patient safety and the efficiency of the pharmaceutical supply chain (Fernandez-Carames & Fraga-Lamas, 2020). Blockchain technology offers a strategic opportunity to overcome these weaknesses by providing a secure, transparent, and decentralized infrastructure. Applications based on blockchain technology have been a major topic of discussion and implementation in the healthcare field (Moreno Arboleda et al., 2024). A blockchain is a distributed digital ledger where data is stored in cryptographically linked blocks, making it highly resistant to tampering or unauthorized modification. In a supply chain context, each product can be given a unique identifier, and its movement is recorded as a series of transactions on the blockchain. This creates a transparent and secure monitoring system for the entire process (Alamsyah & Syahrir, 2023; Tripathi et al., 2023).

The healthcare supply chain is a complex network with several weaknesses, including inefficiency and the circulation of counterfeit drugs, which have caused fatalities primarily among children. Counterfeit drugs can circulate primarily because of the complexity of the healthcare supply chain structure. Therefore, effective monitoring, control, and traceability are fundamental to combating counterfeit drugs within the healthcare supply chain. The study proposed an Ethereum blockchain-based approach that leverages smart contracts and decentralized off-chain storage to enable efficient product tracking within the healthcare supply chain. Testing and validation, as well as cost and security analyses, were carried out to evaluate its effectiveness in improving traceability within the supply chain (Musamih et al., 2021). The application of blockchain to secure the supply chains of drugs, medical devices, and even blood/organs. Blockchain has strong potential to improve traceability, record integrity, and auditability—especially for high-risk products (vaccines, biologics). However, it was emphasized that most of the implementation evidence is still prototypes, lab studies, or small pilots; evidence of long-term economic and operational benefits is limited. also highlighted the issue of interoperability with legacy hospital/distributor systems, as well as the strong need for governance (who writes/validates data) because blockchain does not automatically solve the garbage-in data problem (Fiore et al., 2023). In terms of implementation, the review highlights several case studies in the food, pharmaceutical, and logistics sectors that reported improvements in traceability and reductions in verification time. Nevertheless, the review underscores recurring limitations, particularly regarding scalability (transactions per second), data privacy on-chain, and interoperability challenges. To address issues of data leakage and system performance, the authors recommend a hybrid approach that combines on-chain hashing with off-chain data storage (Liu et al., 2023).

Based on this background, this research aims to develop MediSync, which implements blockchain technology using Hyperledger Fabric in the pharmaceutical supply chain, and is expected to increase transparency and data security. This is especially true regarding the time from when products are produced

until they are received by consumers, to support safer and more effective pharmaceutical services. Using hybrid approach that combines on-chain hashing and off-chain data storage to address data leakage and performance.

2. METHOD

2.1. System architecture

The architecture of the MediSync system is meticulously designed to facilitate secure, transparent, and efficient tracking of pharmaceuticals throughout the supply chain. As depicted in the high-level system architecture in Figure 1, the system is composed of several synergistically interconnected components, each with a distinct role.

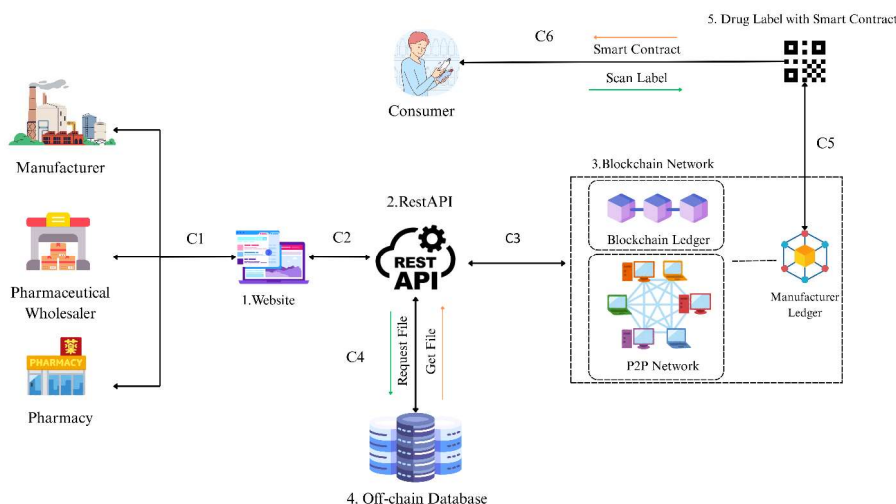


Fig. 1. High Level System Architecture of MediSync

The system's actors are the primary stakeholders in the pharmaceutical supply chain: Manufacturers, who are responsible for the initial production and registration of drug batches; Pharmaceutical Wholesalers (PBF), who manage the distribution from manufacturers to pharmacies; and Pharmacies, who dispense the products to end-users. These actors interact with the system through a Web-Based Interface (Frontend), a modern application built with React. This interface provides a user-friendly dashboard for each role to perform specific tasks, such as creating production schedules, placing orders, or verifying stock, thereby abstracting the underlying complexity of the blockchain operations. All interactions from the front end are processed by the REST API (Backend). This server, built with Node.js and Express, acts as the central nervous system and abstract layer of the platform. It is responsible for handling all incoming HTTP requests, managing user authentication against the off-chain database, and executing the core business logic. Crucially, the backend is the sole intermediary that communicates with both the blockchain network for on-chain transactions and the off-chain database for operational data management.

At the heart of the system is the Blockchain Network, which is implemented using Hyperledger Fabric. This permissioned network maintains a secure, distributed, and immutable ledger shared among all participants. It consists of peer nodes operated by each organization, a Raft-based ordering service that ensures transactional consensus, and a peer-to-peer (P2P) network for communication. This layer serves as the single source of truth for all critical asset information.

To optimize performance and reduce ledger bloat, an Off-Chain Database (MariaDB, managed via XAMPP) is used. This database stores operational data that is either voluminous, frequently changing, or not essential for multi-party consensus, such as user profiles, draft purchase orders, and internal production schedules. It remains seamlessly connected to the blockchain through the REST API, which orchestrates data flow between the two systems.

Finally, the business rules of the consortium are encoded in Smart Contracts (Chaincode). These self-executing contracts automate and enforce processes such as the creation of a new drug batch. For instance, upon successful on-chain registration of a new batch, the backend uses the unique Batch ID returned by the

smart contract to generate a QR Code. This QR code, attached to the physical product, acts as a physical-to-digital bridge, allowing consumers or any stakeholder to scan and retrieve the product's full, immutable history directly from the blockchain, thus ensuring authenticity and transparency from production to consumption.

2.2. Blockchain Network Architecture

The underlying blockchain infrastructure, named medisync network, is a permissioned network implements with Hyperledger Fabric, as depicted in Figure 2.

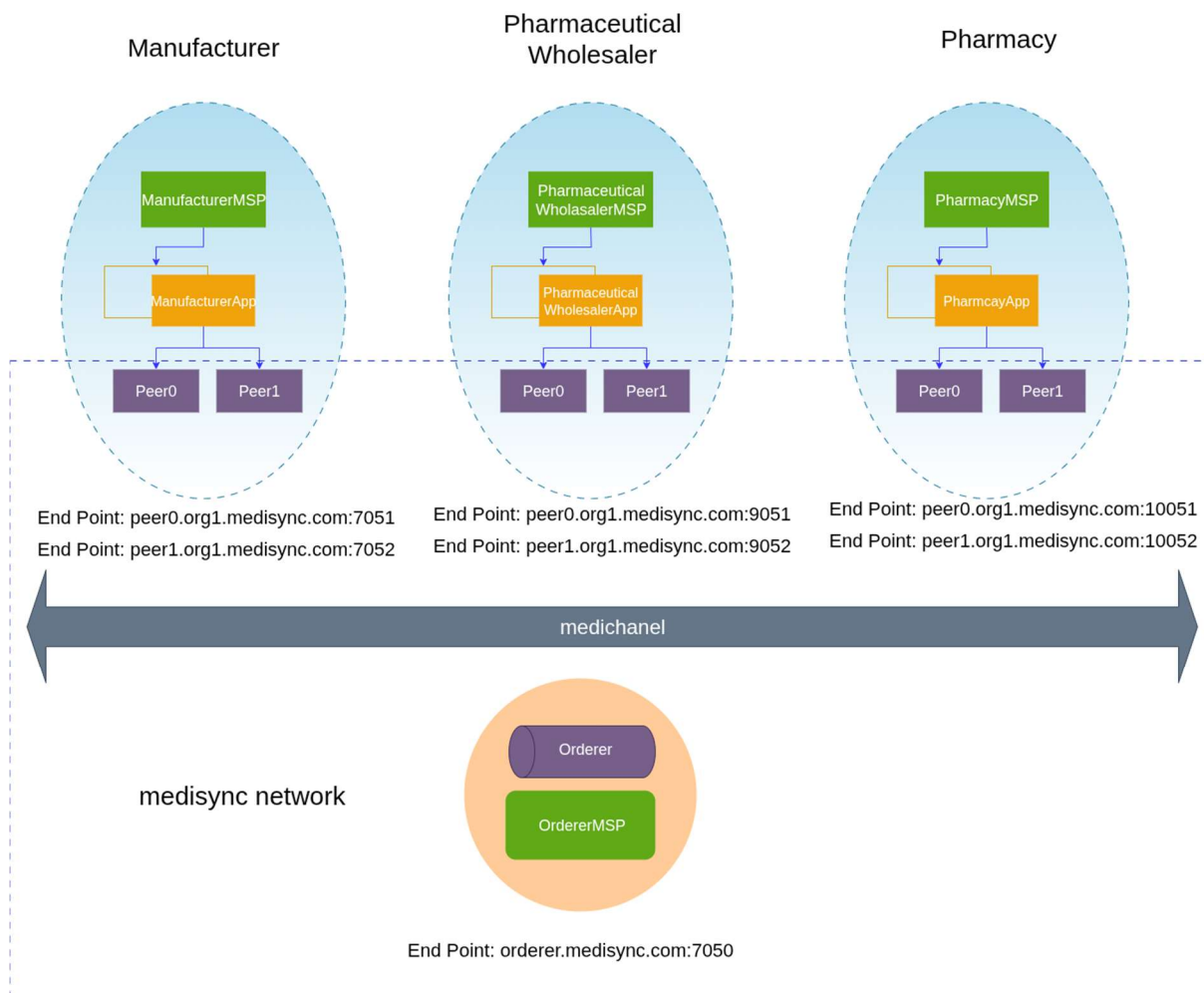


Fig. 2. Blockchain Infrastructure using Hyperledger Fabric

The network consists of three main organizations representing the key stakeholders: Manufacturer, Pharmaceutical Wholesaler, and Pharmacy. Each organization maintains its own peers (e.g., peer0 and peer1), which store a copy of the ledger and executes chain code. Identity and access rights for each organization are managed by a dedicated Membership Service Provider (MSP) and a Certificate Authority (CA). All transactions and communications are conducted within a private channel named medichannel, which ensures data privacy and isolation among the participants. An Ordered service is responsible for sequencing transactions into blocks and distributing them to the peers, ensuring consistent and ordered updates to the ledger across the network.

2.3. Process Flow

The end-to-end business process of the MediSync system ensures a seamless and transparent flow of goods and information, as illustrated in the activity diagram in Figure 3.

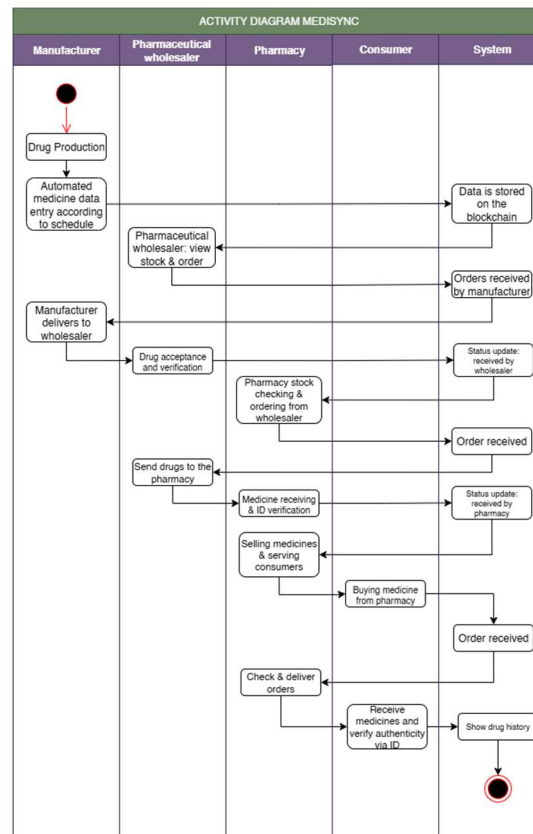


Fig. 3. Activity Diagram of the MediSync Process Flow

The MediSync process flow begins with drug production and data entry, where the manufacturer produces a batch of drugs and records the details into the system. Each batch is assigned a unique ID generated by a smart contract, and the information is automatically stored on the blockchain. In the pharmaceutical wholesaler ordering stage, the wholesaler reviews the available stock and places an order with the manufacturer. This order is recorded as a blockchain transaction and sent to the manufacturer. The manufacturer then ships the drugs to the wholesaler. Upon receipt, the wholesaler verifies the drugs, and the system updates the delivery status on the blockchain on to confirm the successful transfer. The process continues with the pharmacy ordering stage, where the pharmacy checks the wholesaler's stock and submits an order. This transaction is also logged onto the blockchain. The wholesaler then ships the order to the pharmacy. After receiving the drugs, the pharmacy verifies them using their unique ID, and the blockchain is updated to reflect the completed delivery and verification. Finally, in the sale and consumer verification stage, the pharmacy sells the drugs to consumer. Consumer can independently verify the authenticity of the drugs by checking the unique ID against the system, which provides a complete, immutable record of the drug's journey from manufacturer to end user.

3. RESULTS AND DISCUSSION

3.1. Blockchain Network Deployment

The successful implementation of distributed network infrastructure is the first critical result to be verified. The network, designed to represent a consortium of three different organizations (Manufacturer, Pharmaceutical Distributor, and Pharmacy) along with Raft-based ordering services, was launched using Docker Compose. The docker PS command was executed to ensure that all 18 required container components including 6 peer nodes, 6 CouchDB instances, orders, certificate authority, and chain code containers were running correctly. Figure 4 shows a summary output confirming the operational status of the main components.

Fig. 4. Summary output the operational status

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contract function. Upon a successful on-chain transaction, the system generates a unique QR code, as shown in Figure 7.

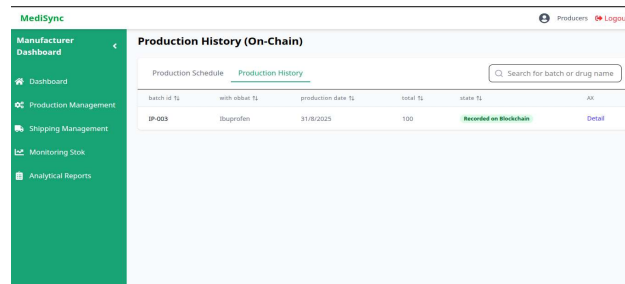


Fig. 6. Production Management page

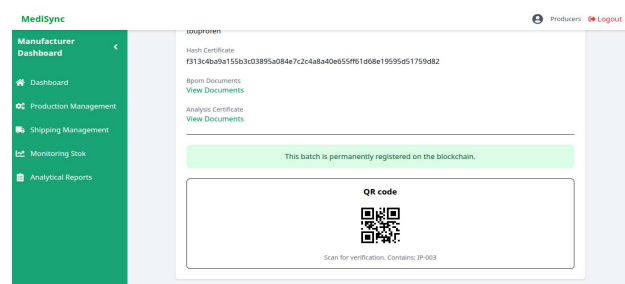


Fig. 7. A successful on-chain transaction with a unique QR code

This end-to-end workflow validates the successful integration of the off-chain operational database with the on-chain auditable ledger. The web application effectively hides the complexities of smart contract invocation from the user. The separation of concerns is a key architectural feature: operational data remains flexible in the off-chain database, while critical, final asset data is committed to the immutable blockchain. The QR code acts as the physical-to-digital bridge, empowering stakeholders to verify a product's authenticity directly from the blockchain, thereby fulfilling a core objective of the mediSync project.

3.4. Validation of Decentralized Governance and Consensus

A fundamental aspect of enterprise blockchain is its model for decentralized governance. In Hyperledger Fabric, this is demonstrated through the chain code lifecycle management process, which requires consensus from consortium members before any business logic can be activated or updated on a channel. To test this, a scenario was simulated where consensus was not achieved.

A new version of the chain code was installed on all peers. Subsequently, an approval transaction was submitted by only one of the three organizations (the Manufacturer). An attempt was then made to commit to the chain code definition with only one out of three required approvals. The network's checkcommitreadiness function correctly identified that only the Producer had approved (ProdusenMSP: true). Consequently, the commit transaction failed, as the channel's default endorsement policy requires most organizations to be not satisfied. The specific error returned by the peer is captured in Figure 8.

```
edged@edi:~/medisync-project$ ./scripts/4.commit.sh
Attempting to commit chaincode v1.0 sequence 1 to the channel...
Checking commit readiness...
{
  "approvals": {
    "ManufacturerMSP": true,
    "PharmaceuticalWholesalerMSP": false,
    "PharmacyMSP": false
  }
}
Committing the chaincode definition...
Error: failed to create signed transaction: proposal response was not successful, error code 500, msg failed to invoke backing implementation of 'CommitChaincodeDefinition': chaincode definition not agreed to by this org (PharmaceuticalWholesalerMSP)
edged@edi:~/medisync-project$
```

Fig. 8. The specific error returned by the peer

This result is crucial as it empirically demonstrates that the system's distributed governance mechanism is functioning correctly. Unlike centralized systems where a single administrator can impose changes, the Fabric

lifecycle ensures that modifications to the shared business logic must achieve consensus. The failed commit proves that the network actively enforces its governance policies, preventing any single organization from unilaterally altering the rules of the consortium. These rules are not mere technicalities; they represent a decentralized voting and ratification process that is securely recorded on the ledger. This ensures that all participants maintain control over the governance of their shared network, a core tenet of enterprise blockchain solutions.

3.5. Public Verification, Traceability via QR Code and Transparency

The core objective of the MediSync platform is to empower external stakeholders, such as consumers and regulators, with the ability to independently verify the authenticity and provenance of a pharmaceutical product. This is achieved through a physical-to-digital bridge implemented via QR codes. Following the successful on-chain registration of a drug batch (as detailed in section 3.3), the system generates a unique QR code containing the product's key identifiers. This QR code is intended for placement on the product's physical packaging. To validate this verification mechanism, a generated QR code was scanned using a standard smartphone camera application (Google Lens). Figure 9 shows Public Verification via QR Code and Figure 10 shows the successful decoding of the QR code consisting of production date, expiry date, and company name.

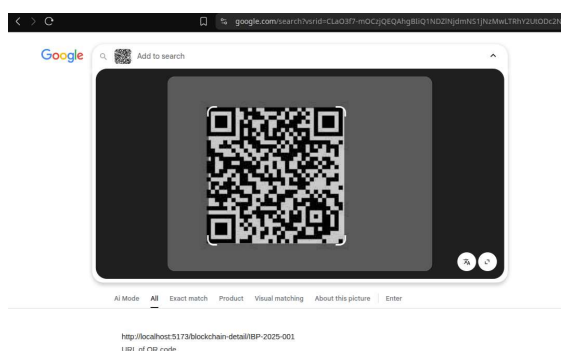


Fig. 9. Public verification via QR Code

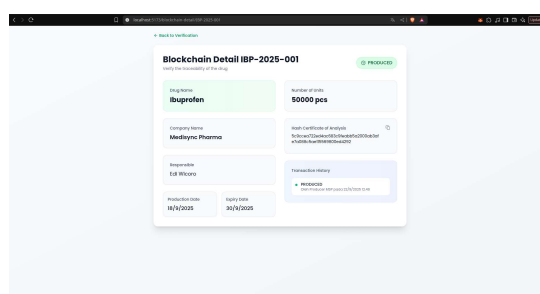


Fig. 10. The successful decoding of the QR code

The scan successfully extracted the embedded JSON payload, which includes the unique batch_id, medicine_name, and expiry_date. This extracted batch_id is then used as the primary key to query the blockchain through a public-facing API endpoint, retrieving the asset's full, immutable history from its genesis. Figures 9 and 10 illustrate how data transparency is achieved through the integration of blockchain with public verification via QR codes. The QR code in Figure 9 serves merely as an entry point, while the actual transparency is demonstrated in Figure 10, where product details, manufacturer information, production volume, certificates of analysis, and transaction history are retrieved directly from the blockchain ledger. This ensures that the public does not merely consume static information but can independently verify the authenticity and integrity of the data. Nevertheless, such transparency remains contingent on the completeness and accuracy of the data initially recorded on the blockchain, which introduces potential bias if the input itself is flawed.

3.6. Data Leakage Testing

In the integrity and data leakage testing of the hybrid architecture (on-chain hash + off-chain storage), we evaluated 1,000 representative data entries, each stored off-chain with its corresponding SHA-256 hash recorded on-chain. The verification results demonstrated 100% consistency between the on-chain hashes and the recalculated hashes from the off-chain files (1,000/1,000), indicating no evidence of inconsistency or detectable data leakage through hash comparison under this test scenario.

Tab. 1. Integrity and data leakage testing of the hybrid architecture

Number of Data Entry	Valid (Hash Match)	Not Valid	Success Rate
1000	1000	0	100%

Although these results demonstrate the effectiveness of on-chain hash recording in detecting data alterations and preventing integrity-based leakage, the results do not yet capture the full spectrum of real-world threats—such as brute-force attacks on small data, metadata exposure, encryption key compromise, or leakage through misconfigured off-chain endpoints. We recommend additional steps before claiming full production-level security, including structured penetration testing of off-chain storage, risk assessment of data reconstruction based on metadata and file size, implementation of end-to-end encryption with robust key management, and load and availability testing to ensure that the verification mechanism remains effective at operational scale. Thus, while the 1,000/1,000 valid outcome provides strong evidence of data integrity under test conditions, further verification against real-world and operational threat vectors is essential to substantiate anti-leakage claims comprehensively.

3.7. Performance Testing

To evaluate the operational efficiency and scalability of the MediSync system, a series of performance tests were conducted using Apache JMeter. The objective was to quantify the performance characteristics of both the off-chain and on-chain components of the hybrid architecture under simulated user load. The tests focused on two primary operations: the initial creation of production data (off-chain) and the subsequent recording of that data onto the blockchain ledger (on-chain).

The results, summarized in Table 2, highlight the distinct performance profiles of each system component. The off-chain system was tested with 1,000 transactions, while the on-chain system was evaluated at various loads, with the 1,000-transaction test presented here for a direct comparison.

Tab. 2. Performance Testing using Apache JMeter

Metric	Off-Chain Result (MySQL)	On-Chain Result (Blockchain)	Analysis
Throughput	101.3 tx/s	7.7 tx/s	Off-chain is approx. 13 times faster
Average Latency	12 ms	1256 ms (1.26 s)	On-chain is approx. 105 times slower
Maximum Latency	75 ms	3087 ms (3.1 s)	On-chain peak latency is significantly higher
Error Rate	0.10%	0.10%	Both systems are highly stable

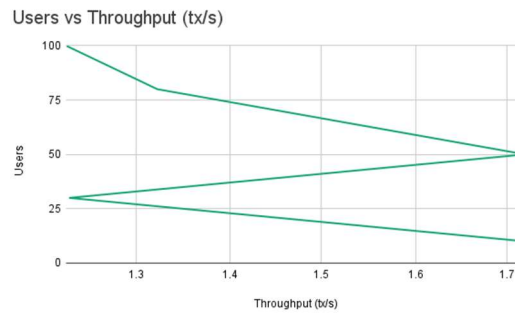


Fig. 11. The graph depicting the relationship between the number of users and throughput

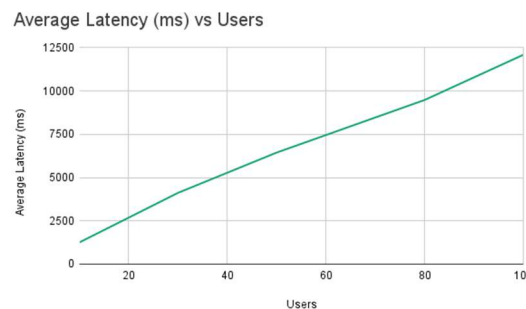


Fig. 12. The graph depicting the relationship between the number of users and latency

The performance testing results using Apache JMeter on the Table 2 revealed a stark contrast between off-chain (MySQL) and on-chain (Blockchain) systems. In terms of throughput, the off-chain system achieved 101.3 transactions per second, while the blockchain recorded only 7.7 transactions per second. This finding highlight that the consensus mechanism in blockchain imposes a serious limitation on processing capacity, making the off-chain system operate approximately thirteen times faster in practice.

The performance gap becomes even more evident in terms of latency. The average latency of the off-chain system was only 12 ms, whereas the blockchain reached 1256 ms (1.26 seconds). In other words, blockchain operates nearly one hundred times slower than the conventional approach such as the performance testing done by Bovenzi and Lau (Bovenzi et al., 2023; Lau et al., 2025). This delay is caused by block validation processes and transaction propagation across all network nodes, which inherently add overhead to the system. At peak conditions, blockchain's maximum latency can increase up to 3087 ms (3.1 seconds), significantly higher compared to 75 ms in the off-chain system.

Interestingly, despite the sharp differences in throughput and latency, the error rate of both systems remained low at 0.10%. This stability confirms that both blockchain and traditional databases are reliable in maintaining transaction consistency. Nevertheless, the graph depicting the relationship between the number of users, throughput, and latency on Figure 11 and Figure 12 exposes a fundamental weakness of blockchain. As the number of users increases, throughput decreases, while latency rises dramatically reaching more than 10 seconds with 100 users. This pattern reflects the inherent limitation of blockchain in handling multi-user scenarios with high interaction rates.

4. CONCLUSION

This research demonstrates that blockchain can substantially improve transparency and data integrity within pharmaceutical supply chains by enabling immutable tracking from production to consumption. The MediSync prototype, built on Hyperledger Fabric, successfully validates decentralized governance, secure auditability, and consumer-level verification through QR codes, thus addressing the pervasive risks of counterfeit drugs and inconsistent reporting systems. However, performance testing underscores critical challenges: blockchain lags significantly behind conventional off-chain systems in both throughput and latency, limiting its viability for high-frequency, real-time applications. These limitations affirm that blockchain should not be viewed as a

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replacement for traditional systems but rather as a complementary layer where data integrity and accountability are paramount. A hybrid model, combining off-chain storage for performance and on-chain hashing for verifiability, emerges as a pragmatic strategy to balance scalability with trustworthiness. Future work should extend to interoperability with legacy healthcare infrastructures, penetration testing of off-chain endpoints, and economic feasibility studies to ensure sustainable adoption beyond prototype environments.

Acknowledgements

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Conflict of Interest

The authors declare that there is no conflict of interest for this manuscript.

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